Short-Read Isoform Quantification as a GLM

Purpose

Common short-read isoform quantification models can be written in a common generalized linear model (GLM) framework. This formulation also defines the read-isoform alignment conditional probability matrix used in quantification-error analysis and K-value calculation.

Basic Model

Consider a gene with T isoforms and relative abundance vector

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_T)'.$$

There are N independent reads $R_1, \ldots, R_N$, each with common read length L. For isoform t, let l_t be the isoform length and

$$\tilde{l}_t = l_t - L + 1$$

be its effective length. Read start sites are assumed to be uniformly distributed over the isoform of origin.

Under this setup, the probability that a read originates from isoform t is

$$\phi_t = \Pr(R_n \in t) = \frac{\theta_t \tilde{l}_t}{\sum_{j=1}^{T} \theta_j \tilde{l}_j}.$$

Read-Based and Region-Based Views

The read-based likelihood models the probability of observing each read directly. A read contributes according to the isoforms to which it can map:

$$\Pr(\text{observing } R_n) = \sum_{t=1}^{T} \frac{1}{\tilde{l}_t} \mathbf{1}\{R_n \text{ maps to isoform } t\} \phi_t.$$

The region/equivalence class-based likelihood first partitions a gene into disjoint regions such that reads falling in the same region map to the same set of isoforms (equivalence classes). If region s has effective length $\tilde{u}_s$, the probability that a randomly selected read falls in region s is

$$p_s = \sum_{t=1}^{T} \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t.$$

Region counts can then be modeled with a multinomial distribution, approximated with independent Poisson variables, or approximated using a normal model for region read proportions.

Main Equivalence Result

In the basic model, read-based and region-based formulations yield equivalent estimators. The key observation is that reads within the same disjoint region have identical mapping compatibility and therefore make the same likelihood contribution. Grouping reads by region turns the read-based likelihood into the multinomial region likelihood up to a constant. The Poisson approximation is also equivalent to these formulations, so the read-based, multinomial, and Poisson likelihoods produce the same maximum-likelihood estimator of $\boldsymbol{\theta}$.

The normal model is not written as the same likelihood, but its mean structure has the same linear form. Thus its estimator approximates the estimators from the read-based, multinomial, and Poisson models.

GLM Formulation

For region read proportions c_s, the normal model uses the identity-link mean equation

$$\mathbb{E}[c_s] = \sum_{t=1}^{T} \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t, \qquad 1 \le s \le S.$$

For region read counts n_s, the multinomial and Poisson formulations use

$$\mathbb{E}[n_s] = N \sum_{t=1}^{T} \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t, \qquad 1 \le s \le S.$$

Dividing the count equation by N gives the same mathematical relationship as the proportion equation. Both are identity-link GLMs relating observed regional summaries to isoform-origin probabilities.

Conditional Probability Matrix

The normal model can be written as a least-squares problem:

$$\hat{\boldsymbol{\phi}} = \arg\min_{\boldsymbol{\phi} \in \mathbb{R}_+^T} \|\mathbf{c} - A\boldsymbol{\phi}\|_2^2 \,,$$

where $\mathbb{R}_+^T$ denotes the nonnegative orthant. Nonnegativity follows from the interpretation of ϕ_t as an isoform-origin probability or nonnegative isoform signal. A sum-to-one constraint is optional: if $\mathbf{c}$ is normalized and the columns of A have compatible total mass, the least-squares loss identifies the scale of $\boldsymbol{\phi}$. The fitted vector can then be normalized post hoc if a probability vector is desired. Here

$$\mathbf{c} = (c_1, c_2, \ldots, c_S)'$$

is the vector of region read proportions and

$$A = (a_{st}) \in \mathbb{R}^{S \times T}$$

is the read-isoform alignment conditional probability matrix. Its entries are

$$a_{st} = \Pr(R_n \in s \mid R_n \in t) = \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\}.$$

Equivalently, the estimation problem is the linear system

$$\mathbf{c} = A\boldsymbol{\phi}.$$

This linear system is the basis for defining the K-value.

Many-Cell Extension and Nuclear-Norm Regularization

For single-cell data, isoforms can be quantified jointly across M cells. The formulation applies either to one gene or to a genome-wide transcript set; the latter is useful when low-rank structure is intended to capture shared cell states rather than gene-specific effects. Let

$$C = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_M] \in \mathbb{R}^{S\times M}$$

be the matrix of region read proportions, with one column per cell, and let

$$\Phi = [\boldsymbol{\phi}_1, \boldsymbol{\phi}_2, \ldots, \boldsymbol{\phi}_M] \in \mathbb{R}^{T\times M}$$

be the matrix of cell-specific isoform signals. The entries of Φ are constrained to be nonnegative. If the same read-isoform compatibility matrix A is used for all cells, the simultaneous least-squares problem is

$$\hat{\Phi} = \arg\min_{\Phi\geq 0} \frac{1}{2} \left\| C - A\Phi \right\|_F^2 .$$

If column-normalized isoform proportions are needed, each fitted column can be converted to

$$\tilde{\boldsymbol{\phi}}_m = \frac{\hat{\boldsymbol{\phi}}_m}{\mathbf{1}'\hat{\boldsymbol{\phi}}_m}$$

after optimization, for columns with positive total mass.

Single-cell read coverage is sparse, so both A and C are typically very sparse. To share information across cells, one can either penalize the nuclear norm of the matrix Φ,

$$\hat{\Phi} = \arg\min_{\Phi\geq 0} \frac{1}{2} \left\| C - A\Phi \right\|_F^2 + \lambda \left\| \Phi \right\|_* .$$

or constrain it directly:

$$\hat{\Phi} = \arg\min_{\Phi\geq 0,\, \|\Phi\|_*\leq \tau} \frac{1}{2} \left\| C - A\Phi \right\|_F^2 .$$

The nuclear norm $\|\Phi\|_*$, the sum of singular values of Φ, favors a low-rank structure in which isoform usage across cells is explained by a small number of shared patterns.

Regularizing Read-Origin or Molecular Abundance

The response matrix is normalized columnwise: for each nonempty cell or pseudobulk, EC counts are divided by their total, so $\mathbf{1}'\mathbf{c}_m = 1$. UMI depth therefore does not set the scale of a column or weight its contribution to the least-squares loss.

The linear model can regularize either read-origin signal Φ or molecular abundance signal Θ, without imposing a simplex constraint. Let $D_l = \mathrm{diag}(\tilde{l}_1, \ldots, \tilde{l}_T)$. Since $\Phi = D_l\Theta$ up to the column scale identified by the normalized response,

$$C \approx A_\phi\Phi = A_\phi D_l\Theta = A_\theta\Theta.$$

Thus regularizing Θ uses $A_\theta = A_\phi D_l$ and replaces $\lambda\|\Phi\|_*$ by $\lambda\|\Theta\|_*$. Under the current geometric approximation, $A_{\phi,st} = \tilde{u}_s/\tilde{l}_t$ and $A_{\theta,st} = \tilde{u}_s$ for compatible isoforms. Fitted columns are normalized only when producing relative-abundance output; the optimization itself remains nonnegative but unconstrained in column sum.

Paired Primer Chemistries

When one biological cell is split into oligo(dT)-primed and random-hexamer- primed half-cells, the two observations should not be treated as independent cells and should not be forced through the same observation model. Let $p \in \{d, h\}$ index the two primers and let A_p be the corresponding EC-by-transcript design. For cell m, each retained half is normalized separately,

$$\boldsymbol{c}_{pm} = \boldsymbol{n}_{pm}/(\mathbf{1}'\boldsymbol{n}_{pm}),$$

and both halves share one coefficient vector:

$$\boldsymbol{c}_{dm} \approx A_d\boldsymbol{\beta}_m, \qquad \boldsymbol{c}_{hm} \approx A_h\boldsymbol{\beta}_m.$$

Here $\boldsymbol{\beta}_m$ is $\boldsymbol{\phi}_m$ or $\boldsymbol{\theta}_m$, with the matching parameterization of each A_p. Consequently, one column of the fitted coefficient matrix represents one biological cell, not one half-cell. The barcode-pair table supplies the mapping from the two observed half-cell columns to that shared fitted column. Pairs are required to be unambiguous within the experimental unit; reused pair keys that cannot identify one biological cell are excluded. The implemented equal-primer-weight objective is

$$\frac{1}{8}\left(\|C_d - A_d\Theta\|_F^2 + \|C_h - A_h\Theta\|_F^2\right) + \mathcal{R}(\Theta),$$

which is evaluated by stacking

$$\widetilde{C} = \frac{1}{2}\begin{bmatrix} C_d \\ C_h \end{bmatrix}, \qquad \widetilde{A} = \frac{1}{2}\begin{bmatrix} A_d \\ A_h \end{bmatrix}.$$

Each column of $\widetilde{C}$ has unit mass. Each column of the stacked A_ϕ also has unit mass; A_θ has the effective-length scaling defined above. The factor of one half fixes the response and design scale used by nuclear-norm regularization; paired-model λ or τ must therefore be selected on this response scale. The current complete-pair implementation requires both halves to pass a primer-specific UMI threshold. A future count-likelihood implementation can retain incomplete pairs by masking the absent primer loss.

The same construction applies to every genome-wide solver below by replacing (C, A) with $(\widetilde{C}, \widetilde{A})$. In particular, direct factorization uses one shared cell-factor row in V for both primer observations. A penalty or constraint is applied once to the shared coefficient matrix, rather than separately to two half-cell estimates. Binary designs use primer-specific compatibility indicators followed by column normalization; fixed weighted designs use the primer-grouped conditional probabilities described next.

The fixed weighted implementation estimates A_d and A_h separately by grouping the patched alevin-fry per-UMI probability sidecar according to the half-cell barcode, averaging likelihood vectors within each primer and EC, and column-normalizing each design. This preserves primer-specific alignment evidence, but the sidecar's hard-coded transcript-end term is shared by both primers and is therefore retained only as an empirical comparator.

The positional-bias design instead classifies each raw Parse read by its RT barcode and streams its transcript read into a primer-specific bulk Salmon run. Exact RT barcodes and unique one-substitution corrections are accepted; unknown or primer-ambiguous barcodes are excluded. To avoid having Salmon's finite bias-training sample come from one large sublibrary, every sublibrary is quantified independently for both primers and the resulting models are combined downstream. The runs use `-l U --posBias --dumpEqWeights`. The explicit unstranded type matches the library type used to generate the alevin EC response; primer-specific auto-detection would otherwise remove ranhex alignments that remain in those counts. Salmon learns separate observed and expected fragment-position distributions in five transcript-length classes and incorporates their ratio into alignment likelihoods and bias-corrected effective lengths.

For primer p, let $q_{pkt}^{(r)}$ be Salmon's dumped rich weight for range-factorized row r, with raw fragment count $n_{pk}^{(r)}$. Rows having the same transcript set are collapsed as

$$\bar{q}_{pkt} = \frac{\sum_r n_{pk}^{(r)} q_{pkt}^{(r)}}{\sum_r n_{pk}^{(r)}}.$$

where r ranges over range-factorized rows and sublibraries. The transcript set, rather than Salmon's local EC number, joins this result to the alevin-fry EC universe. Alevin ECs absent from the Salmon dump receive weights proportional to $1/\tilde{l}_{pt}$ among compatible transcripts, where $\tilde{l}_{pt}$ is that primer's Salmon effective length. Finally each transcript column is normalized over ECs to obtain $A_{\phi,p}$. Join diagnostics report both the fraction of distinct ECs and the primer-specific UMI mass represented by exact Salmon rows. Effective lengths are averaged across sublibraries using mapped fragments as weights.

The sampling transformation must also reflect the primer chemistry. An anchored oligo(dT) reaction gives approximately one UMI-generating priming opportunity per polyadenylated molecule, so its UMI fraction is modeled as proportional to molar abundance (TPM), without a transcript-length factor. Random-hexamer priming can generate more cDNAs from a longer molecule. The primary paired model therefore uses

$$A_{\theta,d} = A_{\phi,d}, \qquad A_{\theta,h} = A_{\phi,h}\,\mathrm{diag}\left(\frac{\tilde{\boldsymbol{l}}_h}{\mathrm{median}_t\, \tilde{l}_{ht}}\right).$$

Median normalization is a global exposure convention: it preserves relative length dependence for random-hexamer UMIs without giving its design an arbitrary base-pair scale relative to the separately normalized oligo(dT) response. The fitted Θ remains shared across the paired halves and is on the oligo(dT) TPM scale. Implemented alternatives apply effective length to both primers or assume TPM-proportional UMIs for both. The selected sampling model and normalization constants are recorded in CV reports and fit diagnostics. Paired-data preparation defaults to the primary oligo(dT)-TPM model.

More exactly, separate normalization of the random-hexamer response gives

$$\mathbb{E}[\boldsymbol{c}_{hm} \mid \boldsymbol{\theta}_m] = \frac{A_{\phi,h} D_h \boldsymbol{\theta}_m}{\mathbf{1}' D_h \boldsymbol{\theta}_m}.$$

The denominator is cell-specific because it depends on that cell's isoform mixture. Replacing it by the median effective length is therefore a linear approximation, not an exact identity. The oligo(dT) block anchors the scale of Θ, because its column-normalized design and response both have unit mass. An exact compositional loss or alternating primer-by-cell exposure updates would account for the denominator, but would no longer be the common linear least-squares problem used by all four solvers.

Sparse Objective Evaluation

A scalable implementation should avoid forming the dense residual $C - A\Phi$. The squared loss can be expanded as

$$\|C - A\Phi\|_F^2 = \|C\|_F^2 - 2\,\mathrm{tr}\big(\Phi' A' C\big) + \mathrm{tr}\big(\Phi' A' A \Phi\big).$$

Thus the data enter through

$$B = A'C, \qquad G = A'A.$$

For a small problem, both can be computed with sparse matrix multiplication. For a genome-wide single-cell matrix, however, B is generally dense even when A and C are sparse, so it must not be materialized. The gradient of the smooth loss is

$$\nabla_\Phi \frac{1}{2} \|C - A\Phi\|_F^2 = G\Phi - B,$$

which requires multiplication by the isoform-by-isoform matrix G, rather than by a large dense region-by-cell residual. When $\Phi = UV'$, with $U \in \mathbb{R}^{T\times r}$ and $V \in \mathbb{R}^{M\times r}$, the required products can be evaluated over cell blocks C_b and V_b:

$$A'C_bV_b, \qquad C_b'AU, \qquad U'GU, \qquad V_b'V_b.$$

Thus the implementation streams sparse cell-by-equivalence-class blocks and keeps only the $T \times r$ and $M \times r$ factors in memory. This avoids forming B, the residual, or the full $T \times M$ abundance matrix during optimization.

Optimization Strategies

There are three practical routes, plus a factorized ADMM variant for the large-scale setting.

- **ADMM with a nuclear-norm split.** Introduce a second matrix Z and solve

  $$\min_{\Phi\geq 0,\, Z} \frac{1}{2} \|C - A\Phi\|_F^2 + \lambda\|Z\|_* \quad \text{subject to} \quad \Phi = Z.$$

  The Φ-update is a sparse nonnegative quadratic problem. The corresponding unconstrained normal equation has the form

  $$(A'A + \rho I)\Phi = A'C + \rho(Z - U),$$

  where U is the scaled dual variable and ρ is the ADMM penalty. Therefore the update can use the sparse products $B = A'C$ and $G = A'A$ instead of forming the dense residual $C - A\Phi$. If T is modest for a gene, factorize $G + \rho I$ once and reuse it for all cells. If T is large, solve the update iteratively using products with sparse A and A', or with G if G is still sparse enough. The nonnegativity constraint can be handled by projected conjugate gradient, coordinate descent, active-set nonnegative least squares, or an additional proximal projection step.

  The Z-update is singular-value soft-thresholding:

  $$Z \leftarrow \mathrm{SVT}_{\lambda/\rho}(\Phi + U).$$

  This step acts on the isoform-by-cell matrix and does not involve A or C. Thus ADMM can exploit sparsity in the data-fitting step, but the nuclear-norm proximal step may become the bottleneck when M is very large. The implemented convex ADMM solver is therefore a bounded reference method: it stores Φ, Z, and the dual variable densely, uses a projected-gradient inner solve for the nonnegative update, and rejects problems beyond a configured dense-entry limit.

  A separate *factorized ADMM* mode replaces the dense state by U, V and nonnegative split copies of these factors. It alternates factor updates with proximal nonnegative projections and stores

$O(r(T+M))$ state. This has the same scalability advantages as the factorized formulation below, but it is non-convex and is not equivalent to the bounded convex ADMM reference.

Cell-factor primal, projected-copy, and dual rows are updated in cached cell blocks. During one epoch the implementation accumulates $V'V$ and CV, then applies A' once and updates the shared transcript-factor primal, copy, and dual. Thus adaptive-ρ updates and primal/dual residuals retain their full-epoch ADMM interpretation; the shared split is not updated from a stochastic cell subset.

Conditional on the transcript factors, the cell-factor primal subproblem has only an r-by-r Hessian. The implementation Cholesky-factorizes this Hessian once per epoch and solves every cached cell block exactly. The transcript-factor gradient and its Lipschitz curvature are both divided by the number of nonempty cells; its step is therefore the reciprocal of the correspondingly scaled curvature. An absolute step cap after this averaging would make optimization slower in direct proportion to the dataset size while allowing the split residuals to become small, so no such cap is used.

Both ADMM implementations can adapt ρ by residual balancing. Every configured interval, let r be the primal split residual and s the dual residual. If $\|r\| > \eta\|s\|$, multiply ρ by κ; if $\|s\| > \eta\|r\|$, divide it by κ. The scaled dual variable is multiplied by the old-to-new ρ ratio so that its unscaled value is unchanged. The implementation defaults to $\eta = 10$, $\kappa = 2$, and an update every ten iterations, and records the complete ρ history.

- **Frank-Wolfe with penalized nonnegativity.** Direct Frank–Wolfe over

$$\{\Phi : \Phi \geq 0,\ \|\Phi\|_* \leq \tau\}$$

does not have the usual singular-vector linear oracle. In particular, replacing the signed descent matrix by its entrywise positive part is not an exact oracle: a nonnegative rank-one outer product can span both positive and negative entries of the original matrix. The legacy `frank_wolfe` implementation makes this approximation and its reported gap is therefore not a valid certificate for the intersection problem.

The scalable corrected variant instead solves

$$\min_{\|\Phi\|_* \leq \tau} \frac{1}{2}\|C - A\Phi\|_F^2 + \frac{\mu}{2}\|\Phi_-\|_F^2, \qquad \Phi_- = \min(\Phi, 0).$$

Its negative gradient is

$$Q_k = B - G\Phi_k - \mu(\Phi_k)_-.$$

The ordinary nuclear-norm ball has the signed rank-one oracle

$$S_k = \tau u_k v_k',$$

where u_k, v_k are leading singular vectors of Q_k. The implementation obtains them with matrix-free power iterations on the signed operator rather than materializing B, Q_k, or Φ_k. Sparse products evaluate the least-squares terms, while transcript-by-cell blocks are reconstructed only to apply $(\Phi_k)_-$.

The update

$$\Phi_{k+1} = (1 - \gamma_k)\Phi_k + \gamma_k S_k$$

preserves the nuclear-norm constraint. A curvature-bounded step uses $\|A(S_k - \Phi_k)\|_F^2 + \mu\|S_k - \Phi_k\|_F^2$. The penalty coefficient is specified relative to the estimated $\|A\|_2^2$, making its scale portable across design matrices. Finite μ gives approximate rather than exact nonnegativity, so manifests report negative Frobenius and absolute-mass fractions. The power-iteration candidate gap is a stopping diagnostic but is explicitly not labeled an exact dual certificate.

- **Low-rank factorization.** Optimize the rank-constrained nonnegative least-squares problem

$$\min_{U\geq 0,\, V\geq 0} \frac{1}{2}\left\|C - AUV'\right\|_F^2, \qquad \operatorname{rank}(UV') \leq r,$$

where $U \in \mathbb{R}^{T\times r}$, $V \in \mathbb{R}^{M\times r}$, and $r \ll \min(T, M)$. This avoids full singular-value decompositions and stores only $O(r(T+M))$ parameters. There is no factor penalty or regularization parameter in this direct method. The loss can be evaluated using B and G:

$$\left\|C - AUV'\right\|_F^2 = \|C\|_F^2 - 2\,\operatorname{tr}\big(VU'B\big) + \operatorname{tr}\big((U'GU)(V'V)\big)\,.$$

The default implementation uses deterministic alternating nonnegative quadratic updates accelerated by FISTA. For fixed U, define

$$H_U = U'A'AU, \qquad K_V = C'AU.$$

The cell-factor gradient is $VH_U - K_V$, and its Lipschitz constant is $L_V = \lambda_{\max}(H_U)$. Starting with $X_0 = V$, $Y_0 = X_0$, and $t_0 = 1$, one projected FISTA step is

$$X_{k+1} = \left[Y_k - L_V^{-1}(Y_k H_U - K_V)\right]_+,$$

$$t_{k+1} = \frac{1+\sqrt{1+4t_k^2}}{2}, \qquad Y_{k+1} = X_{k+1} + \frac{t_k - 1}{t_{k+1}}(X_{k+1} - X_k),$$

where $[\cdot]_+$ is elementwise projection onto the nonnegative orthant. Cells with no retained molecules remain exactly zero.

After updating V, define $H_V = V'V$ and $K_U = A'CV$. The transcript- factor gradient is

$$A'AUH_V - K_U.$$

It uses the same projected FISTA recurrence with the left-acting gradient above and step size

$$L_U^{-1} = \left(1.1\,\|\widehat{A'A}\|_2\,\|H_V\|_2\right)^{-1}.$$

The factor 1.1 guards the matrix-free power estimate of $\|A'A\|_2$. Sparse cell blocks provide $C'AU$ and CV without materializing a dense transcript-by-cell matrix. A configurable number of inner FISTA steps is applied to each subproblem before evaluating the exact sparse objective; the implementation default is 32.

An optional stochastic mode visits every cached cell block once in a seeded random order before deterministic polishing. It is retained for throughput experiments, but is no longer the default because its epoch cost was nearly identical to an exact epoch while its capped, cell-averaged transcript step converged slowly. Saved factors can be validated against cell and transcript identifiers and used as warm starts for deterministic continuation. The patient objective stopping rule is applied only to deterministic epochs: a fit converges after a configured minimum number of epochs followed by a configured number of consecutive relative objective changes below the tolerance. Hitting the iteration ceiling is reported as nonconvergence. Factor columns are rescaled after each epoch to have equal norms while preserving UV', which controls the otherwise arbitrary factor scale without changing the objective. Column normalization is applied after fitting when isoform proportions are required.

The factorized and Frank-Wolfe approaches are the scalable choices for many cells. Both keep Φ in a compact low-rank representation and stream sparse blocks rather than precomputing $B = A'C$. ADMM is useful as a direct convex reference for smaller matrices, but its repeated singular-value thresholding and dense primal state make it unsuitable for a million-cell matrix.

Full-Data Validation

The deterministic FISTA implementation was evaluated genome-wide on a Parse-style single-cell dataset after retaining 169,533 cells with at least 500 raw UMIs. Three-fold molecule-count cross-validation followed an ascending warm-start path over ranks 1, 2, 4, 8, 16, 32, 64, and 128. Every fold/rank fit met the objective-patience stopping rule. Rank 1 failed the profile-variance nondegeneracy criterion for both binary and fixed-weighted designs; all larger ranks passed. Although mean validation loss continued to decrease through rank 128, the one-standard-error rule selected rank 32 for both designs without hitting the rank-grid boundary.

The selected all-cell fits also converged by objective patience. Cell-type labels were not used for rank selection: they were used only afterward as an external evaluation on the 78,552 filtered cells with annotations. The evaluation library-normalizes fitted transcript expression, aggregates it to genes, applies $\log(1 + x)$, computes 30 principal components, and performs group-held-out multinomial prediction. Table 1 summarizes the results.

Design	Rank	Epochs	Objective	Balanced acc.	Macro-F1	Label silhouette
Binary	32	204	1278.199	0.471	0.410	-0.212
Fixed weighted	32	465	690.490	0.386	0.309	-0.305

Table 1: Genome-wide factorized FISTA fits after count-based rank selection.

The paired-primer construction was evaluated separately on 48,568 complete biological-cell pairs, of which 27,383 had reference labels. These fits used rank 64 and shared one coefficient vector across the two primer observations. Both converged, and every labeled fitted profile was active and finite (Table 2). The paired and unpaired rows involve different cell populations, and objectives from binary and weighted designs use different observation matrices, so neither the objectives nor the label scores should be interpreted as controlled head-to-head comparisons.

Design	Rank	Objective	Balanced acc.	Macro-F1	Label silhouette	Labeled cells
Paired binary	64	60.019	0.636	0.567	-0.239	27,383
Paired fixed weighted	64	58.753	0.514	0.439	-0.110	27,383

Table 2: External label evaluation of converged paired-primer fits.

Software Interface and Output

The single-cell command exposes five methods through the `--quant_method` option: `admm`, `admm_factorized`, `frank_wolfe`, `frank_wolfe_penalized`, and `factorized`. The scalable methods use a PyTorch backend and select CUDA when available; NumPy/SciPy remains responsible for sparse input and compatibility-matrix construction. Response rows are normalized once and converted into reusable sparse blocks. The automatic data backend keeps those blocks on the GPU when their estimated storage fits within the configured memory margin; otherwise it keeps pinned host tensors and prefetches them asynchronously. Prepared fold contexts are reused across a complete warm-start path. The rank, cell minibatch size, convergence tolerance, ADMM regularization strength, ADMM penalty, and Frank–Wolfe radius are configurable. The direct factorization rank is selected by count-fold cross-validation; other factorized modes have a configurable rank. The default Frank–Wolfe radius is $\sqrt{M}$. The `--regularization_target` option selects `phi` or `theta`; it changes both the coefficient matrix receiving the penalty and the corresponding design matrix.

In `single_cell` mode, fitting operates directly on the raw cell-by-equivalence-class matrix rather than creating pseudobulks. The output consists of compact low-rank factors, barcode and transcript order, and a manifest of solver diagnostics. Thresholded sparse transcript-by-cell chunks are optional because even sparse reconstruction can be large when many fitted values exceed the threshold. Selected cell blocks can instead be reconstructed from the saved factors. The downstream intron clustering workflow remains a pseudobulk operation and is not invoked in this mode.

Because each response column has unit mass, both random factors are initialized at scale $1/\sqrt{rT}$. This keeps the initial fitted mass at order one; scaling only by rank initializes the fitted mass at order T and can drive a projected method to the all-zero solution on its first update. Projected factor updates use curvature-bounded step sizes rather than one fixed step across datasets. The cell-factor bound is computed from the rank-sized matrix $U'A'AU$. The transcript-factor bound combines $V'V$ with a sparse power-iteration estimate of $\|A\|_2^2$; the ADMM bounds also include the split penalty ρ. This prevents an update from crossing the nonnegative orthant and collapsing an entire factor to zero when the design curvature is large.

The factorized solvers stop only after a configured minimum iteration count and several consecutive relative-objective changes below tolerance. Direct factorization can use noisy minibatch objectives to decide when to begin its deterministic polishing phase, but only polishing iterations establish final convergence. Factorized ADMM additionally requires its relative primal and dual split residuals to satisfy a separately configurable feasibility tolerance. Penalized Frank–Wolfe uses patient relative-objective change because its approximate candidate gap is not a valid certificate. Its atom capacity is configured separately from the fixed rank used by the factorized methods, and additional power iterations can improve the approximate linear oracle. Every manifest records the stopping history, convergence reason, and whether the maximum iteration or atom limit was reached.

For comparison with a standard single-cell expression analysis, downstream label scoring does not apply a nonlinear transform to the arbitrary factors. Instead, transcript loadings are summed by gene so that fitted gene abundance remains VU'_{gene}. For each cell this abundance is normalized to a total of 10,000 and transformed with $\log(1+x)$. Per-fit gene moments are computed in streamed cell blocks, the 2,000 most variable genes are selected, and incremental PCA is fit and transformed in two further streamed passes. Grouped classification, k-means, ARI, NMI, and silhouette are all computed in the resulting 30-dimensional PCA representation. This preserves the intended nonlinear expression preprocessing without materializing a dense genome-wide cell-by-gene matrix.

Choosing the Regularization Level

The implemented cross-validation partitions molecule counts rather than rows of A. Each observed integer EC count is randomly assigned to one of Q folds. For fold q, the remaining counts are normalized per cell and used to fit Φ; the held-out counts are normalized independently and scored with the same design matrix:

$$\mathrm{CV}_q(h) = \frac{1}{M_q} \left\| C_{\text{val}}^{(q)} - A\hat{\Phi}_h^{(q)} \right\|_F^2 .$$

This tests prediction of independent molecules from the same cells while retaining every compatibility pattern in the design and avoiding a separate procedure for unseen cell factors. The workflow can use either a reproducible cell subset or every cell passing a raw molecule-count threshold. It uses three folds, selects a dimensionless multiplier from held-out loss and the representation diagnostics below, and then refits the eligible cells.

Candidate values are dimensionless so that a subset-selected multiplier can be transferred to the complete dataset. For the penalized objective, define

$$\lambda_{\max} = \|A'C\|_2$$

and tune $\lambda/\lambda_{\max}$. For constrained Frank–Wolfe, let u, v be the leading singular vectors of $B = A'C$, with singular value σ, and define the rank-one line-search reference

$$\tau_1 = \frac{\sigma}{\|Au\|_2^2}.$$

The search tunes τ/τ_1. Both scales are estimated by streamed power iterations without materializing B. After selecting a multiplier on the subset, the corresponding scale is recomputed using all cells and their product is passed as the fixed λ or τ for the final fit. The factorized ADMM implementation averages the complete transcript-factor gradient, including its regularization and augmented-Lagrangian terms, by the number of nonempty cells. This preserves the relative scale of the objective terms when a multiplier selected on a subset is transferred to all cells. Within each fold, candidates follow a strong-to-weak regularization path. Factorized ADMM visits decreasing λ and initializes each candidate from the preceding primal factors while resetting the split dual variables. Because the factorized gradient vanishes when both factors are zero, continuation from the $\lambda_{\max}$ solution can otherwise remain at that saddle. For each candidate the implementation also constructs the best rank-one step along the leading singular vectors u, v of $B = A'C$:

$$s_\lambda = \frac{(\sigma - \lambda)_+}{\|Au\|_2^2}, \qquad \Phi_{\text{seed}} = s_\lambda uv'.$$

Small random remaining columns preserve the requested factor rank. The continuation and spectral-seed objectives are evaluated, and the lower one initializes ADMM. The selected all-cell ADMM refit replays this same strong-to-weak construction after recomputing $\lambda_{\max}$ from all eligible cells. Its geometric path includes the closed $\lambda/\lambda_{\max} = 1$ endpoint, a configured smallest positive multiplier, and the exact CV-selected multiplier. The final manifest records the initializer, convergence criterion, residuals, adaptive-ρ state, and objective at every stage; it is therefore possible to verify that the reported endpoint was reached by continuation rather than an unrelated random restart. Frank–Wolfe visits increasing τ; the previous atom factorization is feasible in the enlarged nuclear-norm ball and is retained.

An optimum on an open boundary of the candidate grid triggers automatic geometric expansion. When the new value extends the fitted continuation direction—larger τ for Frank–Wolfe or smaller λ for ADMM—each fold retains its endpoint factors and fits only the new candidate. An expansion that changes the start of a path instead replays that path. Expansion stops at an interior optimum, at zero or $\lambda/\lambda_{\max} = 1$, or at a configured expansion limit; the full fit is rejected only in the last case.

For the penalized Frank–Wolfe solver, model selection uses the one-standard-error rule. Let $\hat{h}$ minimize mean validation loss and let $s_{\hat{h}}$ be its fold standard error. Among converged candidates satisfying $\overline{L}(h) \le \overline{L}(\hat{h}) + s_{\hat{h}}$, the smallest nuclear radius is selected. This favors the most regularized statistically indistinguishable fit and prevents negligible endpoint improvements from driving indefinite grid expansion. The penalized oracle's candidate gap is diagnostic rather than a convergence certificate, so stopping uses patient relative objective change.

Statistical equivalence in reconstruction loss does not make a degenerate representation useful. Before applying the one-standard-error rule, each candidate is therefore screened after rectification, per-cell library normalization, and a log-one-plus transform on a fixed set of high-loading features. Every fold must retain the configured active-cell fraction and exceed a minimum relative between-cell profile variance. This scale-invariant check excludes zero solutions and rank-one factors that vary only by library size.

Among the converged, nondegenerate candidates inside the reconstruction one-standard-error set, define $v(h)$ as mean normalized profile variance. The implementation retains candidates satisfying

$$v(h) \geq \gamma \max_{h'} v(h'),$$

with $\gamma = 0.9$ by default, and chooses the smallest retained nuclear radius. This preserves most observable cell-to-cell structure while retaining the strongest regularization compatible with both structure and held-out count prediction. Cell-type labels remain an external assessment rather than a hyperparameter input.

The direct nonnegative factorization has no λ. Its rank is selected with the same held-out molecule folds. Within each fold the implementation visits candidate ranks from small to large, retains the fitted columns, and adds small positive columns when increasing rank. Among converged, nondegenerate candidates within one standard error of the minimum validation loss, it selects the smallest rank. If that rank is the largest candidate, the grid is doubled up to a configured cap. Each fold retains its largest fitted factorization on the CPU, expands that warm start, and evaluates only the newly added rank rather than replaying ranks that already have held-out scores.

Connecting to equivalence-class-based inference

The preceding derivation used geometric regions with entries $\tilde{u}_s/\tilde{l}_t$. Salmon's range-factorized equivalence classes can be used in the same linear system by replacing those geometric entries with Salmon's conditional assignment weights.

Let K be the number of equivalence classes. Running Salmon with `--dumpEqWeights` provides the two quantities needed to instantiate the model:

- `eq_classes.txt` gives the count c_k for each range-factorized row and Salmon's normalized combined weights

$$q_{kt} \propto \frac{\text{alignment and bias likelihood}_{kt}}{\tilde{l}_t},$$

  with $q_{kt} = 0$ when transcript t is incompatible with class k. These are the coefficients used by Salmon's offline inference, normalized within a row. After duplicate-row collapsing and transcript-column normalization they define the fixed design $A \in \mathbb{R}^{K \times T}$.

- `quant.sf` gives the effective length vector $\tilde{\boldsymbol{l}} = (\tilde{l}_1, \ldots, \tilde{l}_T)'$, including Salmon's fragment-length and bias corrections.

With this substitution, the isoform-origin formulation is exactly the one above: counts use $\mathbb{E}[\boldsymbol{c}] = NA\boldsymbol{\phi}$, or equivalently $\boldsymbol{c} = A\boldsymbol{\phi}^*$ for the unnormalized signal $\boldsymbol{\phi}^* = N\boldsymbol{\phi}$. The same nonnegative least-squares and many-cell extensions therefore apply without explicitly computing geometric region lengths $\tilde{u}_s$.

The binary single-cell implementation constructs A from EC membership when only the standard alevin-fry EC dump is present: compatible entries are weighted by $1/\tilde{l}_t$, using transcript effective-length proxies from the reference, and the region factor is set to one. The patched weighted RAD and alevin-fry route additionally writes per-UMI alignment-likelihood vectors to `gene_eqclass_probs.tsv.gz`. These vectors are computed from alignment score and transcript-end position and are used inside alevin-fry's EM update. They vary by cell and UMI. Using them exactly requires a cell-specific design operator rather than replacing A by a single averaged matrix.

For a direct fixed-design comparison, the implementation supports globally or primer-grouped weighted approximations. For each global EC, its per-UMI likelihood vectors are averaged over the requested cells and UMIs; ECs without dumped vectors receive uniform weights over their compatible transcripts. Each resulting sparse matrix is column-normalized over ECs for each transcript and cached. The binary comparator column-normalizes the binary EC membership matrix in the same way. These comparisons isolate within-EC alignment evidence. They are empirical approximations to the molecule-specific likelihoods, not annotation-only conditional models or complete positional-bias corrections. The paired positional design described above is the implemented primer-specific coverage-bias correction and does not use the sidecar's shared transcript-end prior.

If inference is desired directly on the relative abundance scale $\boldsymbol{\theta}$, absorb transcript effective lengths into the design:

$$B = A\,\mathrm{diag}(\tilde{\boldsymbol{l}}), \qquad b_{kt} = a_{kt}\tilde{l}_t.$$

Solving the corresponding nonnegative system $\boldsymbol{c} = B\boldsymbol{\theta}^*$ estimates an unnormalized abundance signal; normalizing its entries to sum to one recovers relative abundances.